Multimodal Deep Learning

Answer Sheet 6

Date: June 24, 2025

Exercise 1: Information-Theory Basics

(a) **Answer:** Cross-Entropy: $H(p,q) = -\Sigma_{x\in X} p(x)\log q(x)$ is to be minimized, equivalent to minimizing KL Divergence: $D_{KL}(p||q) = \Sigma_{x\in X} p(x)\log \frac{p(x)}{q(x)}$.

(b) **Answer:** By maximizing log-likelihood, $logq_\theta(w)$, will minimize the cross-entropy since $H(p,q_\theta) = -\mathbb{E}_p[\log q_\theta(w)] \propto -\log q_\theta(w)$.

(c) **Answer:** Proof of $H(p,q_\theta) = H(p) + D_{KL}(p||q)$:

$$
\begin{aligned}
H(p,q_\theta) &= -\mathbb{E}_p \log q(x) \\
&= \mathbb{E}_p(\log p(x) - \log p(x) - \log q_\theta(x)) \\
&= -\mathbb{E}_p \log p(x) + \mathbb{E}_p \frac{\log p(x)}{\log q_\theta(x)} \\
&= H(p) + D_{KL}(p||q)
\end{aligned}
$$

(d)